📊 Layoffs Data Analysis Project

Data Cleaning & Exploratory Data Analysis using SQL

🔍 1. Introduction

In recent years, workforce layoffs have become a critical indicator of economic health, business strategy, and market stability.
This project analyzes a **global layoffs dataset** using **SQL** to clean raw data and uncover meaningful insights through **exploratory data analysis (EDA)**.

The focus is to transform messy, real-world data into a **reliable analytical dataset** and answer important business questions such as:

Which industries are most affected?

Which countries faced the highest layoffs?

How layoffs changed over time?

Do well-funded companies also lay off employees?

🎯 2. Project Objectives

Ensure **data accuracy and consistency**

Remove duplicates and irrelevant records

Standardize categorical values

Perform **time-based, company-wise, and industry-wise analysis**

Extract insights useful for business and economic decision-making

🗂️ 3. Dataset Overview

The dataset contains information related to company layoffs with the following attributes:

Column Name	Description
Company	Name of the organization
Location	City/region of operation
Industry	Business sector
Total Laid Off	Number of employees laid off
Percentage Laid Off	Workforce percentage affected
Date	Layoff event date
Stage	Company funding stage
Country	Country of operation
Funds Raised (Millions)	Total funding raised

🧹 4. Data Cleaning Process

✅ 4.1 Duplicate Removal

Used `ROW_NUMBER()` window function to detect duplicate rows.

Retained only the first occurrence of each duplicate record.

📌 *Outcome:*
Eliminated redundancy and improved data integrity.

✅ 4.2 Data Standardization

Trimmed unnecessary spaces from company names

Unified industry labels (e.g., Crypto variations)

Standardized country names (e.g., United States)

📌 *Outcome:*
Improved consistency across categorical fields.

✅ 4.3 Date Formatting

Converted date values from text format to proper DATE datatype.

Enabled accurate year-wise and month-wise analysis.

✅ 4.4 Handling NULL & Blank Values

Removed records where both `total_laid_off` and `percentage_laid_off` were NULL.

Identified missing industry values using self-joins.

📌 *Outcome:*
Final dataset contains only meaningful and analyzable records.

🧾 5. Final Clean Dataset

A cleaned and refined table (`ly6`) was created containing:

Deduplicated records

Standardized text fields

Valid date formats

No irrelevant NULL entries

This dataset serves as a **trusted foundation for analysis**.

📈 6. Exploratory Data Analysis (EDA)

🔹 6.1 Overall Layoff Statistics

Identified **maximum, minimum, and average layoffs**

Highlighted extreme layoff events across companies

🔹 6.2 Company-Wise Layoffs

Calculated total layoffs per company

Ranked companies by workforce reductions

📊 **Insight:**
Large technology firms and fast-growing startups showed the highest layoffs.

🔹 6.3 Industry-Wise Analysis

Aggregated layoffs by industry

📊 **Insight:**
The **Technology and Crypto sectors** were the most impacted.

🔹 6.4 Country-Wise Layoffs

Summed layoffs across countries

📊 **Insight:**
The **United States** experienced the highest number of layoffs, indicating significant market restructuring.

🔹 6.5 Time-Based Trends

📅 Year-Wise Trends

Identified peak layoff years

Observed correlations with economic downturns

📆 Month-Wise Trends

Monthly aggregation revealed sudden spikes

Cumulative layoffs showed long-term impact

📊 **Insight:**
Layoffs intensified during periods of economic uncertainty.

🔹 6.6 Layoffs by Company Stage

Compared layoffs across funding stages

📊 **Insight:**
Both early-stage startups and late-stage companies were affected, showing layoffs are not limited to struggling firms.

🔹 6.7 Funding vs Layoffs

Analyzed layoffs against funds raised

📊 **Insight:**
High funding does not guarantee job security—strategic restructuring plays a major role.

🔹 6.8 Top 5 Companies by Layoffs (Year-Wise)

Used `DENSE_RANK()` to identify the top 5 companies each year

📊 **Insight:**
Different companies dominated layoffs each year, reflecting changing market leaders.

💡 7. Key Insights Summary

✔ Technology sector faced the highest workforce reductions
✔ United States led global layoffs
✔ Layoffs peaked during economic slowdowns
✔ Funding level did not prevent layoffs
✔ Workforce reductions occurred across all company stages

🏁 8. Conclusion

This project demonstrates strong proficiency in **SQL-based data cleaning and exploratory analysis**.
By leveraging **window functions, CTEs, aggregations, and transformations**, raw data was converted into actionable insights.

The findings provide valuable understanding of:

Labor market dynamics

Industry vulnerability

Economic impact on employment

🛠️ 9. Skills & Tools Used

SQL (MySQL)

Data Cleaning & Transformation

Window Functions (`ROW_NUMBER`, `DENSE_RANK`)

CTEs

Exploratory Data Analysis

Business Insight Generation